# SOLVING DYNAMIC MODELS WITH RATIONAL INATTENTION

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**Abstract**

We provide a new algorithm to solve dynamic stochastic general equilibrium models with rational inattention (RI-DSGE). Existing procedures update the equilibrium and attention jointly, through *tâtonnement* on the aggregate paths. We decouple the two: an inner loop solves the general equilibrium given attention, and an outer loop updates attention given the equilibrium. The inner loop relies on an exact mapping of attention into expectation matrices, under which the equilibrium is a single linear system in the sequence space. Our algorithm thus converges in far fewer iterations, which allows the attention problems to be solved on an almost exact representation of the equilibrium dynamics. Applied to models from the literature, it reaches the same equilibria as the published procedures up to 1,100 times faster, with the largest gains where existing procedures must take small steps to remain stable. The speed gains make likelihood-based estimation of RI-DSGE models feasible.

**Keywords:** rational inattention, information frictions, expectations, DSGE, computational methods.

**JEL Codes**: C61, C63, D83, D84, E32, E52

[†] The toolbox implementing the method will be made public in the future; in the meantime, it is available upon request.

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# 1. Introduction

Dynamic stochastic general equilibrium (DSGE) models are one of the main macroeconomic frameworks for quantitative research and policy analysis. Considerable effort has gone into making solution methods for such models accessible, to the point that the computational burden is often no longer a concern (see Adjemian et al. 2011 and Auclert et al. 2021 for widely used toolboxes). Most of these methods can be modified to accommodate departures from perfect information rational expectations (Mankiw and Reis, 2007; Gabaix, 2020; Angeletos and Huo, 2021), but only with the degree of information frictions taken as given; limited attention enters the model like any other exogenous source of sluggish adjustment.

Sims (1998)'s *rational inattention* (RI) instead derives the degree of information frictions from a microfounded problem: limited attention is an equilibrium outcome that responds to policy and the economic environment (see Maćkowiak and Wiederholt, 2015, 2025 and Afrouzi and Yang 2021). RI pairs particularly well with linearized DSGE models, in which optimal attention takes the form of Gaussian signals about the state of the economy (Maćkowiak, Matějka, and Wiederholt, 2018). In expectation, actions satisfy the same optimality conditions of the perfect information economy, and the Kalman filter computes these expectations from each agent's information set. The drawback is that solving RI-DSGE models takes considerable know-how: it requires finding "the fixed point of an economy in which the optimal signal of an agent depends on the signals chosen by other agents" (Maćkowiak and Wiederholt 2025, p. 274), for which no *one-shot* solution method exists.

In this paper, we propose a *new* algorithm for solving RI-DSGE models that is substantially faster than existing procedures, fast enough to solve the agents' attention problems on an almost exact representation of the equilibrium dynamics rather than on a low-order approximation of it. Existing procedures revise a guess of the aggregate paths jointly with attention, a form of *tâtonnement*, so general equilibrium feedbacks—such as strategic complementarities or interactions between agents' attention choices—are resolved only across iterations. Our algorithm separates the two. Given attention, an inner loop solves exactly for the paths of all endogenous variables in the sequence space. The outer loop then updates attention given the equilibrium paths.[1]

How our method departs from existing procedures is best understood by analogy with value function iteration and policy iteration (Howard, 1960), two methods for solving Bellman equations. Value function iteration applies the Bellman oper-

[1]We damp the update of attention; the paths could be damped instead, with attention updated fully at each iteration, and the fixed point is the same (Section 4).

Table 1: Solution Time

| | | Time (seconds) | | | Contraction rate $\rho$ | |
|---|---|---|---|---|---|---|
| Model | Shock | Published | Ours | Speed-up | Published | Ours |
| AY21 | Monetary & technology | 34.6 | 17.2 | 2.0 | 0.982 | 0.965 |
| MW15 | Monetary | 714 | 4.7 | 152 | 0.989 | 0.694 |
| | Technology | 95.8 | 3.8 | 25 | 0.989 | 0.690 |
| MW25 | Technology, $h = 0$ | 5,582 | 5.1 | 1,094 | 0.9999 | 0.578 |
| | Technology, $h = 2$ | 5,197 | 32.4 | 160 | 0.979 | 0.815 |

*Notes:* In MW25, the technology shock is anticipated $h$ periods ahead ($h = 0$: no news). Published damping steps are the authors'; on MW15 and MW25, raising them to 1.5 or 2 times their value makes the published procedure at most 1.7 times faster, or makes it diverge. For ours, we pick the fastest of a few candidates. The contraction rate $\rho$ is the factor by which the distance to the solution shrinks per iteration, estimated from the last 50 iterations. Hardware: AMD EPYC 9655, 4 cores; Julia 1.10.

ator and simultaneously updates the value and the policy at every step, so both converge together, often slowly. Policy iteration instead evaluates the value of the current policy exactly before improving it, and typically converges in far fewer iterations. Current RI-DSGE solvers resemble value function iteration: each iteration takes aggregate paths as given and updates both attention and the equilibrium. Our algorithm resembles policy iteration: the inner loop solves the equilibrium implied by the current attention exactly before the outer loop updates attention. Like existing procedures, it is guess-and-verify—but over attention only—with each candidate checked against the exact equilibrium it implies.

In the sequence space, the equilibrium given attention is a single linear system that delivers the paths of all endogenous variables at once. To obtain the system, we map attention into expectation matrices, extending Auclert, Rognlie, and Straub (2020)'s treatment of sticky expectations to rational inattention: given the solution of the attention problems, the mapping follows from the average filtered estimate of the history of innovations. Combined with the perfect information optimality conditions, these matrices then deliver the actions under rational inattention. Existing procedures instead compute the perfect information actions given the guessed paths, and then pass them to the attention problems. In state space, an exact solution would require a model-specific state that includes the agents' average estimates. Our representation needs no such augmentation and gives access to the other tools of the sequence space, such as the representation of models as directed acyclic graphs

of blocks (Auclert et al., 2021).

Decoupling attention from the general equilibrium also makes our method more robust. In existing procedures, an error in the guessed paths carries directly into the updated ones, amplified by general equilibrium feedbacks; when these feedbacks are strong, the iteration is stable only with very small steps. Over the many iterations it then needs, small numerical errors can accumulate, whether from solving the attention problems or from the compact representations of the paths used to keep each iteration cheap.[2] In our procedure, the general equilibrium is solved exactly at each iteration, so an error in the paths survives only through the attention choices it induces. In the models of Maćkowiak and Wiederholt (2015) and Maćkowiak and Wiederholt (2025), this channel is weak: in the direction that dominates convergence, an undamped iteration of our procedure leaves at most 40% of the error, against 90% for the published procedure in Maćkowiak and Wiederholt (2015), while in Maćkowiak and Wiederholt (2025) the published procedure overshoots the solution many times over in the direction that limits its step (Section 5). Our procedure can therefore take large steps and converge in far fewer iterations, which makes it affordable to solve the attention problems on the almost exact MA($T$) representation, where $T$ is the truncation horizon of the sequences.

We demonstrate the effectiveness of our method by solving RI-DSGE models from the literature.[3] Table 1 reports the solution times: our algorithm reaches the same equilibrium 2 to 1,094 times faster than the published methods. The gains are largest where existing procedures must take small steps to remain stable, as in Maćkowiak and Wiederholt (2025), and smallest in Afrouzi and Yang (2021), where strategic complementarity in price setting slows down both procedures alike. The increased speed renders likelihood-based estimation feasible on a single machine, which we illustrate on the model of Maćkowiak and Wiederholt (2015): each likelihood evaluation, which solves the model, takes about 18 seconds, so that a chain of 5,000 draws runs in 25 hours, against roughly 47 days at the solution times of the published procedure.

**RI-DSGE models in the literature.** Maćkowiak and Wiederholt (2009) embed inattention in the price-setting framework of Woodford (2003), in which nominal aggregate demand follows an exogenous process. Maćkowiak and Wiederholt (2015) build a small-scale DSGE model wherein households and firms are both inattentive. Afrouzi and Yang (2021) return to inattentive price setters, but with an explicit

[2]Fitting persistent dynamics with low-order VARMA processes often produces complex roots, which introduce spurious oscillatory components.

[3]Not to be confused with models in which attention is chosen outside the general equilibrium, such as Pfäuti (2023).

demand side, closed by an interest rate rule whose changes they study. Song and Stern (2025) split firms into attentive and inattentive types facing monetary policy shocks. Maćkowiak and Wiederholt (2025) build a real business cycle model in which inattentive firms and households face technology shocks that can be anticipated. All of these models are solved by iterating on the paths of aggregate variables.[4]

Within each of these models, the attention problems are solved either by numerical optimization or, more efficiently, by iterating on their first-order conditions (Afrouzi and Yang, 2021; Miao, Wu, and Young, 2022). We propose a way to accelerate these iterative procedures on large state spaces, which may be of independent interest.

**Layout.** The remainder of the paper is structured as follows. Section 2 represents the equilibrium, given the expectation matrices, as a single linear system in the sequence space. Section 3 formulates the attention problem and shows how its solution maps into the expectation matrices. Section 4 describes the algorithm. Section 5 applies it to RI-DSGE models from the literature and shows that its speed makes likelihood-based estimation feasible. Section 6 concludes.

## 2. Equilibrium Representation

We consider economies whose equilibrium can be written as a system of linear stochastic difference equations.[5] Fluctuations arise from a vector of $n_\varepsilon$ structural innovations $\varepsilon_t$, Gaussian and i.i.d. over time. Collecting the $n_x$ endogenous variables in $x_t$, the equilibrium conditions read:

$$E_t\left[\Phi_+ x_{t+1} + \Phi_0 x_t + \Phi_- x_{t-1} + \Phi_\varepsilon \varepsilon_t\right] = 0, \tag{1}$$

and the solution of the model expresses the endogenous variables as a function of the state variables—their own lags and the innovations:

$$x_t = \Pi_- x_{t-1} + \Pi_\varepsilon \varepsilon_t. \tag{2}$$

Several well-known methods solve such systems under perfect information and rational expectations (Blanchard and Kahn 1980, Klein 2000, Sims 2002). Under imperfect information, the state must also carry the agents' beliefs, and when agents learn from endogenous variables—so that the forecasts of others matter—the equi-

[4] Afrouzi and Yang (2021) also work in the sequence space, but still through *tâtonnement*.

[5] Typically in log-deviations from the non-stochastic steady state; i.e., $x_t := \log X_t - \log X$.

librium may admit no finite-state representation such as Equation (2) (Townsend, 1983).[6]

To keep the state independent of the information structure, we work instead with the moving-average representation of the equilibrium presented in Section 2.1, whose state is the truncated history of innovations.[7] Information frictions then alter only the mapping from the sequences of the agents' targets to those of their actions, given by the expectation matrices of Section 2.2.

**2.1. Sequence Space.** A causal linear equilibrium driven by the structural innovations can be expressed as a linear function of current and past innovations:

$$x_t = \sum_{\tau=0}^{T-1} \psi_\tau^x \, \varepsilon_{t-\tau}, \tag{3}$$

where $\psi_\tau^x$ is the response of $x_t$ to the innovations that occurred $\tau$ periods earlier. The truncation at horizon $T$, the only approximation of the representation, is chosen so that the responses have converged. Beyond $T$, stationary responses are set to zero and permanent ones are held at their value at $T$. We stack the entries of $\psi_0^x, \ldots, \psi_{T-1}^x$ into one vector, the *sequence* of $x$, denoted $\boldsymbol{x}$, which collects the impulse responses of $x$ to every innovation at every horizon. Throughout, bold symbols denote sequences and the matrices that act on them, and plain symbols denote variables and coefficients at a given date. Under perfect information rational expectations, $\psi_\tau^x = \Pi_-^\tau \Pi_\varepsilon$, and the sequence of a variable is also, by certainty equivalence, its perfect foresight path following an innovation at date 0 that no one anticipated—an "MIT" shock.

Since each equilibrium condition is linear, it holds separately for each innovation and each horizon, and the equilibrium can be written as a linear relation between the sequences of the variables it involves:

$$\boldsymbol{\Phi}_x \, \boldsymbol{x} + \boldsymbol{\Phi}_\varepsilon \, \boldsymbol{\varepsilon} = \mathbf{0}, \tag{4}$$

where $\boldsymbol{\varepsilon}$ is the sequence of the innovations themselves—a unit impulse at horizon 0 for each innovation—and $\boldsymbol{\Phi}_x$ and $\boldsymbol{\Phi}_\varepsilon$ stack $\Phi_-$, $\Phi_0$, $\Phi_+$ and $\Phi_\varepsilon$ horizon by horizon and innovation by innovation. Equation (4) is the sequence-space representation of Auclert et al. (2021), a stacked system in the tradition of Anderson and Moore (1985).

[6] Nimark (2007) obtains a finite-state approximation by truncating the hierarchy of higher-order expectations included in the state.

[7] Lorenzoni (2009) similarly truncates a history, of aggregate variables rather than innovations, and filters on the truncated state.

**2.2. Expectations.** Some of the variables in $x_t$ are chosen by inattentive agents, that is, agents who face information frictions. These agents belong to groups indexed by $g$, such as firms or households, and each group is populated by a continuum of agents $\iota \in [0,1]$. Agents in group $g$ share the same optimality conditions, which define the group's *target* $x^*_{gt}$ (e.g., the profit-maximizing price for a firm, or the consumption implied by the Euler equation for a permanent-income household), the action they would take under perfect information, given the aggregate paths. Each agent sets its action to its best estimate of the target given its information set $\mathcal{I}_{\iota gt}$:

$$x_{\iota gt} = E\left[x^*_{gt} \mid \mathcal{I}_{\iota gt}\right]. \tag{5}$$

Only the cross-sectional average of the actions within each group, $\bar{x}_{gt} := \int_0^1 x_{\iota gt}\, d\iota$, enters the vector of endogenous variables $x_t$.

We focus on information sets consisting of Gaussian signals about the current and past state of the economy, with noise that is idiosyncratic across agents. Section 3 describes these signals and how agents choose them. Each agent's estimate of the target is then linear in the innovations and in its signal noise, and the noise integrates to zero over the continuum, so the sequence of the average action is a linear function of the target's sequence, given by a matrix $\boldsymbol{\mathcal{E}}_g$:

$$\bar{\boldsymbol{x}}_g = \boldsymbol{\mathcal{E}}_g\, \boldsymbol{x}^*_g. \tag{6}$$

We call $\boldsymbol{\mathcal{E}}_g$ the *expectation matrix* of group $g$, following Auclert, Rognlie, and Straub (2020), who introduce it in the context of sticky expectations. Consider first the case of a single structural shock, $n_\varepsilon = 1$, and an innovation at date 0. Row $h$ of $\boldsymbol{\mathcal{E}}_g$ collects the average estimates held at date $h$: its entry in column $s$ is the average estimate, at date $h$, of an innovation at date $h - s$. The diagonal entry of row $h$, in column $s = h$, is thus how much of the actual innovation agents have learned by date $h$, and the other entries of the row how much of it they attribute to innovations at other dates. Under perfect information the expectation matrix is the identity; if agents learn nothing it is zero; if they eventually learn the innovation, the diagonal entry of row $h$ tends to one and the other entries of the row to zero as $h$ grows. When the group chooses several actions, all of them are based on these same estimates, so the same matrix applies to each component of the target.

Expectation matrices also accommodate the confounding of structural shocks when $n_\varepsilon > 1$: a signal that does not separate the shocks (on their sum, for instance) leads agents to attribute part of one innovation to another. Since the sequences are stacked as vectors across innovations and horizons, the expectation matrix is then made of $n_\varepsilon \times n_\varepsilon$ blocks of size $T \times T$, one for each pair of innovations. Each block

has the structure of the single-shock case: the block pairing an actual innovation with a second one collects the average estimates of the second following the first. Without confounding, the off-diagonal blocks are zero and each diagonal block is the expectation matrix of a single shock; with confounding, the off-diagonal blocks measure how much of each innovation is attributed to the others.

The sequence-space representation and the expectation matrices together make up the inner loop of our algorithm: given the expectation matrices, the equilibrium is a single linear system in the sequences, solved at once for the paths of all endogenous variables. The next section describes how, given these paths, the attention problems are solved and the expectation matrices obtained.

## 3. Attention

This section lays out the microfoundations of the expectation matrices, and with them the outer loop of our algorithm: given the equilibrium paths, each agent solves an attention problem that determines its signals about the state, and the Kalman filter maps these signals into the expectation matrix of the group. Since the attention problem has the same form for every group, we drop the group index $g$ throughout.

**3.1. Signals.** The state of the economy is the history of innovations, which evolves by shifting one period and adding the new innovations:

$$\xi_t := \begin{bmatrix} \varepsilon_t \\ \vdots \\ \varepsilon_{t-T+1} \end{bmatrix}, \qquad \xi_{t+1} = F\xi_t + \begin{bmatrix} \varepsilon_{t+1} \\ 0 \end{bmatrix}. \tag{7}$$

Every variable, the target included, is then a linear function of $\xi_t$, with its impulse responses Equation (3) as loadings:[8]

$$x_t^* = \Psi^* \xi_t, \qquad \Psi^* := \begin{bmatrix} \psi_0^{x^*} & \cdots & \psi_{T-1}^{x^*} \end{bmatrix}. \tag{8}$$

Agents receive noisy signals about the state:

$$S_{\iota t} = \Gamma\, \xi_t + \nu_{\iota t}, \qquad \nu_{\iota t} \sim \mathcal{N}(0, \Sigma_\nu), \tag{9}$$

where the noise $\nu_{\iota t}$ is i.i.d., independent of the innovations, and idiosyncratic, so that it integrates to zero over the group. Signals of this form are optimal under the

[8]The forecasts of other agents enter the target through these loadings. When a target's response converges to a nonzero level, as the price level's does, the state also carries the sum of the innovations older than $T$, on which the target loads with that long-run response.

attention problem of Section 3.2 (see Maćkowiak, Matějka, and Wiederholt 2018). A signal on a variable—a price, output, the interest rate, last period's sales—is a signal on $\xi_t$ with the impulse responses of that variable as loadings, which is why $\Gamma$ depends on the equilibrium. The signals can be specified exogenously, in which case $\Gamma$ and $\Sigma_\nu$ are given and the expectation matrix follows directly from Section 3.3, or chosen by the agents, as under rational inattention.

**3.2. Dynamic LQG-RI Problem.** Under rational inattention, attention is costly: each agent designs its signals recognizing that processing more information reduces the expected losses from suboptimal actions but comes at a higher attention cost. We formulate the problem as a dynamic linear-quadratic-Gaussian rational inattention (LQG-RI) problem. In period $t = -1$, agent $\iota$ chooses an attention strategy $\{\Gamma, \Sigma_\nu\}$ to solve:[9]

$$\max_{\{\Gamma,\Sigma_\nu\}} \left\{ \sum_{t=0}^{\infty} \beta^t E_{\iota,-1} \left[ \frac{1}{2} (x_{\iota t} - x_t^*)' \Theta (x_{\iota t} - x_t^*) \right] - \lambda \sum_{t=0}^{\infty} \beta^t I(\xi_t; S_{\iota t} \mid \mathcal{I}_{\iota,t-1}) \right\}, \quad (10)$$

subject to the law of motion Equation (7), the signal equation Equation (9), the mapping Equation (8) from $\xi_t$ to the target $x_t^*$, and the optimality condition $x_{\iota t} = E[x_t^* \mid \mathcal{I}_{\iota t}]$. The matrix $\Theta$, negative definite, weighs the losses incurred from the deviation of the actions from their targets; when the problem is microfounded, $\Theta$ is the Hessian of the agent's objective with respect to its actions, evaluated at the non-stochastic steady state. The scalar $\lambda > 0$ is the marginal cost of attention, and $I(\xi_t; S_{\iota t} \mid \mathcal{I}_{\iota,t-1})$ is the mutual information between the state and the signal conditional on past information, which measures the per-period reduction in uncertainty.

Afrouzi and Yang (2021) characterize the solution: the optimal signals load on the directions of the state in which the marginal benefit of information—the gain from tracking the target more closely today plus the value of a more precise prior tomorrow—exceeds its cost, and their number never exceeds the number of actions. The attention strategy therefore depends on the equilibrium only through the mapping from the state to the target, $\Psi^*$. The problem can be posed with attention paid to all shocks jointly, or to each shock as a distinct activity. On the history of innovations, the state has $n_\varepsilon T$ coordinates, which makes the standard iteration on the first-order condition costly; Appendix A.1 describes how we solve the problem at that scale.

[9] We focus on the steady-state solution of the problem, in which the uncertainty about the state, conditional on past information, has converged, so that the attention strategy and the Kalman gain are time-invariant; a single pair, rather than a sequence of pairs indexed by time, then describes the strategy.

**3.3. Mapping Signals to Expectation Matrices.** Given the signals, the estimate $\hat{\xi}_{\iota t} := E[\xi_t \mid \mathcal{I}_{\iota t}]$ follows from the Kalman filter:

$$\hat{\xi}_{\iota t} = F\hat{\xi}_{\iota,t-1} + K\,\varsigma_{\iota t}, \qquad \varsigma_{\iota t} := S_{\iota t} - \Gamma F\hat{\xi}_{\iota,t-1}, \tag{11}$$

with $\varsigma_{\iota t}$ the surprise in the signal and $K$ the steady-state gain, whose expression and Riccati recursion are given in Appendix A.2. Since the state is the history of innovations, the filtered estimate contains the estimate of every past innovation, and the filter delivers their revisions over time. Averaging Equation (11) across the group, the noise integrates to zero, and the average estimate $\bar{\xi}_t$ follows:

$$\bar{\xi}_t = (I - K\Gamma)F\,\bar{\xi}_{t-1} + K\Gamma\,\xi_t. \tag{12}$$

The expectation matrix is obtained by feeding an innovation at date 0 into Equation (12) and iterating it for $T$ periods, starting from $\bar{\xi}_{-1} = 0$: the average estimate $\bar{\xi}_h$ gives row $h$ of the expectation matrix, with its entries ordered as in Section 2.2.[10] Given the signals, the expectation matrix does not depend on the target, and applies to whatever target the inner loop produces.

The history of innovations is the most transparent state and, in practice, the most accurate one, but not the cheapest. When a low-order VARMA representation fits the targets closely, the attention problem is computationally easier to solve on its state-space realization, which has a handful of coordinates instead of $n_\varepsilon T$. Appendix A.3 describes this compact state and how the estimates of past innovations are recovered on it.

## 4. Algorithm

This section states the equilibrium as a fixed point between the sequences and the expectation matrices, describes our algorithm and the existing procedures in the same notation, and discusses how to run the algorithm in practice.

**4.1. The Fixed Point.** Given the expectation matrices $\boldsymbol{\mathcal{E}} := (\boldsymbol{\mathcal{E}}_g)_g$, the equilibrium is the linear system of the inner loop:

$$\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}})\,\boldsymbol{x} + \boldsymbol{\Phi}_\varepsilon\boldsymbol{\varepsilon} = \mathbf{0}, \tag{13}$$

where $\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}})$ is $\boldsymbol{\Phi}_x$ with each group's optimality conditions replaced by Equation (6). The expectation matrices, however, depend on the equilibrium, $\boldsymbol{\mathcal{E}} = \boldsymbol{\mathcal{E}}(\boldsymbol{x})$, so that

[10] With several shocks, each entry becomes a block of $n_\varepsilon$ estimates, and the recursion is run once for each innovation.

an equilibrium is a fixed point:

$$\boldsymbol{\Phi}_x\big(\boldsymbol{\mathcal{E}}(\boldsymbol{x})\big)\,\boldsymbol{x} + \boldsymbol{\Phi}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0}. \tag{14}$$

Two maps are involved. Given the sequences, the map $\boldsymbol{x} \mapsto \boldsymbol{\mathcal{E}}(\boldsymbol{x})$ solves each group's attention problem and builds its expectation matrix from the chosen signals (Section 3); this map is non-linear—the inherent difficulty of rational inattention, under which the signals themselves adjust to the equilibrium. Given the expectation matrices, the map $\boldsymbol{\mathcal{E}} \mapsto \boldsymbol{x}$ is a linear solve. The fixed point can be stated in either object, and the choice of the object iterated on leaves the equilibrium unchanged.

Convergence can likewise be measured on either object. We measure it on the aggregate variables that enter the agents' problems, which every procedure produces, so that the same criterion applies to ours and to the published ones (Section 5). Existing procedures represent these paths either by truncated impulse responses or by VARMA processes refitted at every iteration; with the latter, convergence is sometimes checked on the coefficients instead.

**4.2. Iterating on Expectation Matrices.** Our algorithm iterates on the expectation matrices and solves the equilibrium exactly in between:

1. Start from initial expectation matrices $\boldsymbol{\mathcal{E}}^{(0)}$, and solve $\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}}^{(0)})\boldsymbol{x}^{(0)} + \boldsymbol{\Phi}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0}$.
2. Given $\boldsymbol{x}^{(n)}$, solve each group's attention problem and build its expectation matrix: $\tilde{\boldsymbol{\mathcal{E}}}^{(n+1)} = \boldsymbol{\mathcal{E}}(\boldsymbol{x}^{(n)})$.
3. Damp: $\boldsymbol{\mathcal{E}}^{(n+1)} = (1-\omega)\boldsymbol{\mathcal{E}}^{(n)} + \omega\tilde{\boldsymbol{\mathcal{E}}}^{(n+1)}$, with $\omega > 0$.[11]
4. Solve $\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}}^{(n+1)})\boldsymbol{x}^{(n+1)} + \boldsymbol{\Phi}_\varepsilon \boldsymbol{\varepsilon} = \mathbf{0}$, and return to step 2 until convergence.[12]

The part of $\boldsymbol{\Phi}_x(\boldsymbol{\mathcal{E}})$ that does not involve the expectation matrices is built once; only the rows of the groups' conditions change across iterations.

**4.3. Iterating on Aggregate Paths.** Existing procedures instead iterate on the aggregate paths that agents take as given. In our notation, they run as follows:

1. Guess the dynamics of the aggregate variables entering the targets, $\boldsymbol{x}^{(0)}$.

[11] The admissible range of $\omega$ depends on the strength of the feedback of the paths on the expectation matrices; values $\omega > 1$ over-relax the update.

[12] Convergence is declared when the largest change in the aggregate variables that enter the agents' problems, between the current paths and those implied by the undamped update of the expectation matrices, falls below a tolerance.

2. Given $\boldsymbol{x}^{(n)}$, solve each group's attention problem and build its expectation matrix: $\boldsymbol{\mathcal{E}}(\boldsymbol{x}^{(n)})$.

3. Set the average actions of each group to $\boldsymbol{\mathcal{E}}_g(\boldsymbol{x}^{(n)})\,\boldsymbol{x}^*_g(\boldsymbol{x}^{(n)})$, with the targets evaluated at the guess, and obtain the remaining variables from the other equilibrium conditions given those actions: $\tilde{\boldsymbol{x}}^{(n+1)}$.

4. Damp: $\boldsymbol{x}^{(n+1)} = (1-\omega)\boldsymbol{x}^{(n)} + \omega\tilde{\boldsymbol{x}}^{(n+1)}$, and return to step 2 until convergence.

Step 2 is common to both procedures; the difference lies in step 3. Because the targets are evaluated at the guess, actions and targets are never solved jointly: the equilibrium feedback—from the price level to the firms' target and back to the price level, say—is worked out across iterations, through the damping of the guess. In our algorithm, the same feedback is resolved within each iteration, and only the expectation matrices are damped.

Where the feedback is resolved matters for convergence. In the *tâtonnement*, the damping must contract the feedback of the equilibrium on itself, which is often strong—strategic complementarity in price setting, the response of the wage to labor demand—so the step has to be small and convergence takes many iterations. In our algorithm, the damping only has to contract the feedback of the paths on the expectation matrices, which is typically weak by comparison: a change in the paths moves the agents' estimates of the innovations much less than it moves their targets. Larger steps, and far fewer iterations, therefore suffice. Each of our iterations can cost more, since it solves the full equilibrium, but the larger steps more than make up for it (Section 5).

**4.4. Practical Choices.** Nothing in the iteration requires starting from perfect information, $\boldsymbol{\mathcal{E}}^{(0)} = \boldsymbol{I}$. Initializing the algorithm from the solution of a nearby calibration saves iterations: with information costs 5% higher, 67 to 85% fewer in the models of Table 1. With several groups of inattentive agents and when the initial guess is far from the fixed point solution, it can be preferable to solve in two stages: first with a single inattentive group and then with all of them, starting the second stage from the solution of the first.

A few further choices can speed up the algorithm. When the targets admit a low-order VARMA representation, the attention problems can be solved on its state-space realization, a much smaller state than the history of innovations (Appendix A.3). Each attention problem can be warm-started from the prior covariance, the marginal benefit of information and the attended subspace of the previous outer iteration, which are then close to their fixed point (Appendix A.1). The attention problems should be solved to a tighter tolerance than the outer loop, so that their numerical

error does not stall it (we use $10^{-7}$ and $10^{-5}$). Finally, the steps of both the outer loop and the attention problems can be tuned, for instance by picking the fastest of a few candidates, and the outer loop can be further accelerated by standard fixed-point methods such as Anderson acceleration (Anderson, 1965).

## 5. Applications

We now take the algorithm to RI-DSGE models from the literature: Section 5.1 compares its computational cost with that of the published procedures on the same models, and Section 5.2 uses it to estimate a Maćkowiak and Wiederholt (2015) model with Bayesian methods. In Section 5.1, every procedure stops when the largest change in the aggregate variables that enter the agents' problems falls below $10^{-5}$, between the current paths and those implied by the undamped update, of the expectation matrices in ours and of the paths in the *tâtonnement*, with the attention problems themselves solved to a tolerance of $10^{-7}$.

**5.1. Solving Models from the Literature.** We solve the models of Afrouzi and Yang (2021), Maćkowiak and Wiederholt (2015) and Maćkowiak and Wiederholt (2025), described in Appendix B, with their published procedures and with ours, holding fixed the calibration, except for Maćkowiak and Wiederholt (2015)'s costs of attention, which we halve (Table 5), and the state of the attention problems used in the original papers.[13] All procedures solve the attention problems with the same solver, at the same inner step and tolerance, so the differences in Table 2 come from the outer loop alone. At each calibration, all procedures reach the same solution, with responses within 0.05% of their peak. Against the authors' own solutions, the largest gap is 0.3% of the peak in Afrouzi and Yang (2021), 1.5% in Maćkowiak and Wiederholt (2025) (on its first stage, the one the authors store) and, at the halved costs, 28 to 44% in Maćkowiak and Wiederholt (2015).

At the authors' step, our procedure needs fewer iterations in most models (Table 2, Panel A): 2.5 times fewer in Afrouzi and Yang (2021), about 11 times fewer in Maćkowiak and Wiederholt (2015), and 1.8 times fewer in Maćkowiak and Wiederholt (2025) with news; without news, it needs 17% more. In Maćkowiak and Wiederholt (2025), the steps of $2 \times 10^{-4}$ and $5 \times 10^{-4}$ require 57,000 to 117,000 iterations under either procedure, and since each of our iterations is costlier, the gain in time is small or negative: our procedure is faster with news and slower without. The main gain comes from the step size. Our procedure remains stable at steps of 0.2 to 0.5, at

[13] When the attention problems are solved on the state-space realization of a VARMA representation of the targets, the expectation matrices are recovered as described in Appendix A.3.

which it converges in 35 to 130 iterations and at most 33 seconds in Maćkowiak and Wiederholt (2015) and Maćkowiak and Wiederholt (2025), 25 to almost 1,100 times faster than the published procedures.[14]

Figure 1 shows where the gain comes from. Each line traces, iteration by iteration, how far a procedure still is from convergence. At the authors' step, both procedures close in slowly: each iteration removes only a small share of the remaining error, so the lines descend gently and reach the tolerance after hundreds to more than a hundred thousand iterations. At larger steps, our procedure drops by several orders of magnitude within a few dozen iterations. Panel B of Table 2 summarizes these paths by the contraction rate $\rho$, the share of the error left after one iteration, and by the feedback $\mu$, the share left after one undamped iteration in the direction that dominates at that step (Appendix C). In the *tâtonnement*, the feedback is either close to one, as in Maćkowiak and Wiederholt (2015), where an undamped iteration leaves about 90% of the error, or strongly negative in the direction that limits the step, as in Maćkowiak and Wiederholt (2025), where the divergence at slightly larger steps implies that an undamped iteration overshoots the solution by thousands of times the error. The published procedures therefore need small steps, and with small steps their contraction rates are close to one. In our procedure, an error survives only through the agents' attention choices: in Maćkowiak and Wiederholt (2015) and Maćkowiak and Wiederholt (2025), the feedback of the slowest direction lies between 0.1 and 0.4, and a step of 0.5 removes 30 to 40% of the error at each iteration; with news in Maćkowiak and Wiederholt (2025), another direction keeps our step below 0.5. Afrouzi and Yang (2021) is the exception: the feedback is close to one under both procedures, 0.98 and 0.97, because the strategic complementarity in price setting carries an error from one iteration to the next however the equilibrium is solved. The gain there is accordingly a factor of 2 rather than of 25 or more.

**5.2. Likelihood-Based Estimation.** The speed of our procedure makes it possible to estimate RI-DSGE models by full-information methods, which require solving the model at every evaluation of the likelihood. We illustrate with Maćkowiak and Wiederholt (2015)'s model, in which firms set prices and households choose consumption and set wages under rational inattention, with technology and monetary policy shocks. Each agent allocates attention separately to the two shocks. We simulate 200 quarters of output and inflation from the model at a known calibration,

[14]Larger steps gain little. At 1.5 and 2 times the published step, the published procedure is at most 1.7 times faster on Maćkowiak and Wiederholt (2015), and it diverges at 3 times the step with the technology shock, the one case we ran at that step. On Maćkowiak and Wiederholt (2025), it is 1.4 times faster at 1.5 times the step without news, and it diverges at twice the step without news and at 1.5 times the step with news.

Table 2: Iterations, Solution Time and Contraction

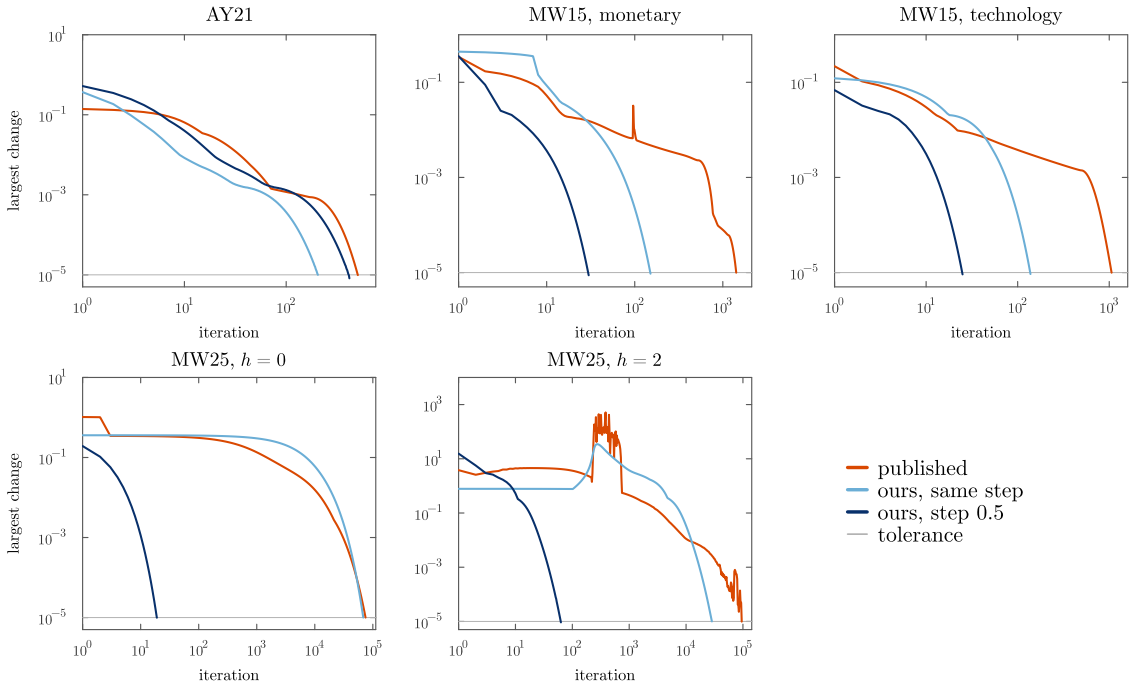
| | | | Published | | Ours, same step | | Ours, step 0.5 | |
|---|---|---|---|---|---|---|---|---|
| *Panel A: Iterations and time* | | | | | | | | |
| Model | Shock | Step | Iter. | Time | Iter. | Time | Iter. | Time |
| AY21 | Monetary & technology | 1 | 507 | 34.6 | 206 | 17.2 | 423 | 30.3 |
| MW15 | Monetary | 0.1 | 3,598 | 714 | 307 | 20.3 | 59 | 4.7 |
| | Technology | 0.1 | 3,112 | 95.8 | 287 | 19.1 | 53 | 3.8 |
| MW25 | Technology, $h = 0$ | $2{\cdot}10^{-4}$ | 100,449 | 5,582 | 117,052 | 10,217 | 35 | 5.1 |
| | Technology, $h = 2$ | $5{\cdot}10^{-4}$ | 105,188 | 5,197 | 57,030 | 4,415 | 130 | 32.4 |
| *Panel B: Contraction rate and feedback* | | | | | | | | |
| Model | Shock | Step | $\rho$ | $\mu$ | $\rho$ | $\mu$ | $\rho$ | $\mu$ |
| AY21 | Monetary & technology | 1 | 0.982 | 0.98 | 0.965 | 0.97 | 0.981 | 0.96 |
| MW15 | Monetary | 0.1 | 0.989 | 0.89 | 0.940 | 0.40 | 0.694 | 0.39 |
| | Technology | 0.1 | 0.989 | 0.89 | 0.928 | 0.28 | 0.690 | 0.38 |
| MW25 | Technology, $h = 0$ | $2{\cdot}10^{-4}$ | 0.99990 | $0.52^a$ | 0.99986 | 0.28 | 0.578 | 0.16 |
| | Technology, $h = 2$ | $5{\cdot}10^{-4}$ | 0.979 | $-40^a$ | 0.99964 | 0.27 | 0.815 | 0.08 |

*Notes:* Time in seconds. In MW25, the technology shock is anticipated $h$ periods ahead ($h = 0$: no news). In MW15 and MW25, where firms and households are both inattentive, we solve in two stages, first with inattentive firms only, then with both; Panel A sums the iterations over the two. For MW25 with $h = 2$, the largest admissible step is below 0.5; we use 0.2. Panel B: $\rho$ is the contraction rate per iteration at the step used, and $\mu = 1 - (1 - \rho)/\omega$ the feedback, for step $\omega$ (Appendix C); both are estimated from the last 50 iterations of the stage in which firms and households are both inattentive (AY21: firms). [a] These values take the direction that dominates the distance at the end of the iteration not to oscillate. For $h = 0$, a larger step supports this: at 1.5 times the step, $\rho$ falls to 0.99981, whereas an oscillating direction with the same $\rho$ would diverge. For $h = 2$, the distances do not settle it, and the same $\rho$ is also consistent with $\mu \approx -4{,}000$. In both cases, the small published steps reflect strongly negative feedback in some direction: the iteration diverges at twice the published step for $h = 0$ and at 1.5 times it for $h = 2$, which implies a most negative feedback of about $-5{,}000$ to $-6{,}700$ and $-2{,}700$ to $-4{,}000$, respectively.

which differs from that of Table 5: the true values of the estimated parameters are in Table 3, with costs of attention per nat on responses in percent, and the others are $\beta = 0.99$, $\gamma = 1$, $\psi = 1$, $\varphi = 2/3$, $\theta = 6$ and $\theta_w = 6$. We then estimate the model on these data.

The likelihood is exact up to the truncation of the impulse responses. The sequence-space solution delivers the model's impulse responses directly; truncated at a horizon of 100 quarters, where at the true parameters they are below $10^{-4}$ of

Figure 1: Convergence of the Outer Iterations




*Notes:* The published procedure runs at the authors' step; ours at the same step and at a step of 0.5. In MW15 and MW25, the panels show the stage in which firms and households are both inattentive; in AY21, the single stage with inattentive firms. For MW25 with $h = 2$, the largest admissible step is below 0.5; we use 0.2. The grey line is the tolerance, $10^{-5}$.

their peak, they give the autocovariances of the observables at every lag, and hence their joint Gaussian density (Auclert et al., 2021, Sections 5.2–5.3). With two shocks and two observables, this density is well defined. We estimate two parameter sets: the two costs of attention and the interest-rate smoothing; and, in addition, the coefficients of the Taylor rule and the persistence and volatility of the two shocks, nine parameters in all. The priors take the distribution families of Herbst and Schorfheide (2016) for the standard parameters and are log-normal for the costs of attention.

We locate the posterior mode with the covariance matrix adaptation evolution strategy (CMA-ES; Hansen and Ostermeier, 2001), the most reliable of the optimizers Andreasen (2010) compares on DSGE likelihoods. We then run a random-walk Metropolis–Hastings chain of 5,000 draws, the first 1,000 discarded. The proposal is proportional to the covariance of CMA-ES's final search distribution, which approximates the shape of the inverse Hessian (Hansen, 2016); it is scaled during the burn-in toward an acceptance rate of 25%. At each draw, the model is solved on the

exact state, the window of the last 100 innovations. Rather than starting from perfect information, each solve starts from expectation matrices close to the solution: those at the posterior mode. A separate run of the three-parameter chain solves every candidate both from this start and from the expectation matrices of the last accepted draw. The two starts give the same log posterior, to within 0.04 over 700 candidates, and take about the same time per solve. A draw takes 17.8 seconds with three parameters and 21.3 with nine, so a chain runs in 25 and 30 hours on a single machine. As a rough extrapolation, at the solution times of the published procedure in Table 2, 810 seconds for the two shocks at the calibration and horizon of Table 5, the same chain would take about 47 days.[15] The chains here serve as an illustration. Tighter posteriors take more draws, and since the cost of any sampler is the number of solves times the time per solve, our speed gains carry over directly. Each solve is independent of the others, so the draws can also be spread across a cluster, or a sequential Monte Carlo sampler can evaluate many particles in parallel and update them online as new data arrive (Cai et al., 2021).

Table 3 reports the posterior medians and 90% intervals, with the time per draw and the acceptance rate. With three parameters, all three intervals contain the true values. With nine parameters, all nine do, but the intervals on the three shared parameters are wider, most of all for the firms' cost of attention, and the data say little about $\phi_\pi$, $\phi_y$ and $\rho_m$, whose posteriors stay close to their priors.

**Potential pitfalls.** Strategic complementarities in information acquisition *can* generate multiple equilibria, with attention choices reinforcing each other (Hellwig and Veldkamp, 2009; Yang, 2015). These tend to arise when attention falls below a certain model-specific threshold.[16] If the posterior puts mass near that threshold, the equilibrium a draw reaches can depend on its starting point. Started from the last accepted draw's solution, a proposal may drift to a low-attention equilibrium that an initial guess from a more attentive region would avoid. The equilibrium then depends on the path of the chain. We therefore start every draw from the same point, the posterior mode, which implicitly defines a selection rule; the comparison of the two starts above covers the three-parameter chain only, and during the mode search each

[15] Each solve in the chain stops when the largest change falls below $10^{-4}$ with the attention problems solved to $10^{-7}$. We did not tune the algorithm for the estimation: tuning the steps or accelerating the outer loop (Section 4.4) could reduce the time per solve further. Conversely, the published procedures would also take fewer iterations if started from the solution at a nearby draw, so the comparison overstates the gain in that respect.

[16] Multiple equilibria can be detected by solving the model from many initial expectation matrices and checking whether all of them converge to the same equilibrium. Zero attention tests directly for a low-attention equilibrium, and for estimation the solutions at neighboring parameter values matter most.

Table 3: Estimation on Simulated Data

| | | Three parameters | | Nine parameters | |
|---|---|---|---|---|---|
| Parameter | Truth | Median | 90% interval | Median | 90% interval |
| Firms' cost of attention $\lambda_f$ | 120 | 107.8 | [92.5, 126.5] | 131.0 | [65.8, 222.5] |
| Households' cost of attention $\lambda_h$ | 30 | 28.4 | [22.0, 37.7] | 29.1 | [19.3, 49.0] |
| Interest-rate smoothing $\rho_r$ | 0.9 | 0.897 | [0.885, 0.910] | 0.890 | [0.799, 0.931] |
| Taylor rule, inflation $\phi_\pi$ | 1.5 | – | – | 1.46 | [1.16, 1.87] |
| Taylor rule, output $\phi_y$ | 0.125 | – | – | 0.054 | [0.012, 0.224] |
| Technology persistence $\rho_a$ | 0.9 | – | – | 0.895 | [0.855, 0.932] |
| Technology volatility $\sigma_a$ | 0.85 | – | – | 0.876 | [0.638, 1.085] |
| Monetary persistence $\rho_m$ | 0.1 | – | – | 0.191 | [0.046, 0.401] |
| Monetary volatility $\sigma_m$ | 0.2 | – | – | 0.163 | [0.089, 0.454] |
| Seconds per draw | | 17.8 | | 21.3 | |
| Acceptance rate | | 0.20 | | 0.20 | |
| Total time (hours) | | 24.8 | | 29.6 | |

*Notes:* 200 simulated quarters of output and inflation. Exact likelihood. 4,000 draws after a burn-in of 1,000. Each proposal is initialized from the solution at the posterior mode. Costs of attention are per nat, on responses in percent.

evaluation starts from the solution at the best point so far. We view this as no more of a concern than the common practice, in perfect information rational expectations models, of restricting estimation to the determinacy region, which rules out sunspot equilibria.

## 6. Conclusion

We propose a new algorithm for solving RI-DSGE models that decouples attention from the general equilibrium. Given attention, an inner loop solves exactly for the paths of all endogenous variables in the sequence space; the outer loop then updates attention given those paths. Because general equilibrium feedbacks are resolved within each iteration rather than across iterations, the outer loop tolerates large steps and converges in far fewer iterations, which in turn makes it affordable to solve the attention problems on an almost exact representation of the equilibrium dynamics. On the models of Afrouzi and Yang (2021), Maćkowiak and Wiederholt (2015) and Maćkowiak and Wiederholt (2025) our algorithm reaches the same equilibrium as the published procedures 2 to 1,094 times faster.

RI-DSGE models have so far remained confined to a handful of papers, arguably not for lack of interest: "many economists find the idea of rational inattention plau-

sible" (Maćkowiak, Matějka, and Wiederholt, 2017, p. 4), but due to the difficulty of finding the fixed point between attention choices and the equilibrium dynamics they induce. DSGE models themselves largely spread through academia and policy institutions once software made them tractable and portable (Cherrier, Saïdi, and Sergi, 2023). Our toolbox aims to do the same for RI-DSGE models. Models with several groups of inattentive agents and rich dynamics over long horizons can now be solved in seconds rather than hours, fast enough for likelihood-based estimation. In Briand (2026), we apply our method to a medium-scale RI-DSGE model with capital accumulation and price setting.

From a broader perspective, our method broadens the use of expectation matrices in the sequence space. Auclert, Rognlie, and Straub (2020) use them with exogenous information frictions; we take them to economies in which agents learn from endogenous variables, so that the forecasts of others matter and the expectation matrices themselves depend on the equilibrium. Rational inattention is one instance of such economies; the algorithm applies whenever agents filter Gaussian signals about the endogenous state, whether these signals are specified exogenously or derived from another model of information choice.

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# A. Attention: Technical Details

**A.1. Solving LQG-RI on Large States** With Gaussian signals, the mutual information in Equation (10) equals $\frac{1}{2}\left[\log\det\Sigma_- - \log\det\Sigma\right]$, where $\Sigma_-$ and $\Sigma$ are the steady-state prior and posterior covariances of the state (Appendix A.2). The posterior solves a first-order condition in which the marginal benefit of information is the gain from tracking the target more closely today—a quadratic form in the state, given by $\Theta$ and the mapping Equation (8)—plus the discounted value of a more precise prior tomorrow (Afrouzi and Yang, 2021). Writing $\Omega$ for this benefit, the optimal signals are on the eigenvectors of:

$$\Sigma_-^{1/2}\,\Omega\,\Sigma_-^{1/2} \tag{15}$$

whose eigenvalues exceed $\lambda/2$, with noise chosen so that the corresponding eigenvalues of $\Sigma^{1/2}\Omega\Sigma^{1/2}$ equal $\lambda/2$; there are at most as many such signals as actions, $n_a$ (their Theorem 2.2).

We solve this first-order condition with the fixed-point map of Afrouzi and Yang (2021), which iterates on the prior covariance and $\Omega$. On the history of innovations, the state has $n := n_\varepsilon T$ coordinates, and each step of the map requires two eigendecompositions of a dense $n \times n$ matrix: one for the square root of the prior, one for Equation (15). We avoid both. First, any square root of the prior will do: with its Cholesky factor $C$, the matrix $C'\Omega C$ has the same eigenvalues as Equation (15), and its attended eigenvectors $u$ give the same posterior, with signal loadings $C^{-\prime}u$ and reductions of the covariance along $Cu$. Second, only the attended eigenpairs matter: we find the top $n_a + 1$ by subspace iteration on $v \mapsto C'\Omega(Cv)$, without forming the matrix and starting from the subspace found at the previous step of the map; the $(n_a+1)$-th computed eigenvalue falling below $\lambda/2$ indicates that no attended direction is missed, which the agreement in Table 4 supports on our examples. A step then costs one Cholesky factorization, $O(n^3)$, and a subspace iteration, $O(n_a n^2)$, and the map converges to the same fixed point. Table 4 compares the two on three problems of Maćkowiak, Matějka, and Wiederholt (2018) and on one with two targets.[17]

When the attention problems are solved many times, as in our outer loop or in estimation, we recommend warm-starting each one from the prior covariance, $\Omega$ and attended subspace of the previous solution, which is then close to its fixed point.

**A.2. The Filter** In steady state, the gain of the Kalman filter Equation (11) is:

$$K = \Sigma_-\Gamma'\left(\Gamma\Sigma_-\Gamma' + \Sigma_\nu\right)^{-1}, \tag{16}$$

[17]The four problems are included as examples in our toolbox.

Table 4: Attention Problems: Solution Time

| Problem | State | $n$ | Time (seconds) AY21 | Ours | Speed-up |
|---|---|---|---|---|---|
| AR(2) | MA(100) | 101 | 29.5 | 1.23 | 24 |
| | MA(200) | 201 | 114 | 2.88 | 39 |
| | MA(400) | 401 | 561 | 6.69 | 84 |
| Price level | MA(100) | 101 | 3.47 | 0.054 | 64 |
| | MA(200) | 201 | 24.6 | 0.179 | 137 |
| | MA(400) | 401 | 186 | 0.762 | 243 |
| Two targets | MA(100) | 202 | 6.63 | 0.105 | 63 |
| | MA(200) | 402 | 19.6 | 0.410 | 48 |
| | MA(400) | 802 | 83.6 | 1.94 | 43 |
| News | MA(100) | 101 | 1.46 | 0.155 | 9.4 |
| | MA(200) | 201 | 4.74 | 0.461 | 10 |
| | MA(400) | 401 | 19.8 | 1.67 | 12 |

*Notes:* Each row is one attention problem with given targets: three examples of Maćkowiak, Matějka, and Wiederholt (2018)—an AR(2) target, price setting (the state has a unit root), and a target that responds to a shock announced 8 periods ahead—and a problem with two targets and two shocks. MA($T$) is the history of the last $T$ innovations; $n$ is the dimension of the state. AY21 is the fixed-point map of Afrouzi and Yang (2021); both maps run at the same step, tolerance and iteration cap, and agree to the fifth decimal of the information flow. Hardware: AMD EPYC 9655, 4 cores; Julia 1.10.

where the prior covariance $\Sigma_- := \mathrm{Var}(\xi_t \mid \mathcal{I}_{\iota,t-1})$ and the posterior covariance $\Sigma := \mathrm{Var}(\xi_t \mid \mathcal{I}_{\iota t})$ are the fixed point of the Riccati equations:

$$\Sigma = \Sigma_- - K\Gamma\Sigma_-, \qquad \Sigma_- = F\Sigma F' + \mathrm{diag}\left(\Sigma_\varepsilon, 0, \ldots, 0\right). \tag{17}$$

**A.3. Compact States** The attention problem can instead be solved on a compact state when a low-order VARMA representation fits the targets closely. Its state-space realization, $\zeta_{t+1} = A\zeta_t + B\varepsilon_{t+1}$, has a handful of coordinates instead of $n_\varepsilon T$, which makes the attention problem much cheaper, and the expectation matrices—hence the sequence-space solution of the inner loop—remain available. The compact state retains only part of the history of innovations, so the estimates of past innovations follow instead from their projection on the surprises observed since they occurred:

$$E[\varepsilon_\tau \mid \mathcal{I}_{\iota t}] = \sum_{d=\tau}^{t} P_{d-\tau}\, \varsigma_{\iota d}, \qquad P_\ell = \Sigma_\varepsilon B' \left[(A(I - K\Gamma))'\right]^\ell \Gamma'\Sigma_\varsigma^{-1}, \tag{18}$$

with $\Sigma_\varsigma := \Gamma\Sigma_-\Gamma' + \Sigma_\nu$ the variance of the surprise. When no low-order VARMA fits the sequences, or when the order required makes refitting at every iteration costlier than it saves, as with news shocks announced many periods ahead, the attention problem is solved directly on the history of innovations.[18]

## B. Models

All variables are in log-deviations from the non-stochastic steady state. Rationally inattentive agents set their action to the conditional expectation of their target, with the same losses as in the original papers. Leads in the targets and the equilibrium conditions are expectations at date $t$; along the impulse responses, once the innovation is known, they coincide with the future values, which is how we write them. Table 5 reports the calibrations. For each model, we list the endogenous variables entering the targets, on which convergence is measured.

**B.1. Afrouzi-Yang 2021.** Firms set prices under rational inattention, and households have perfect information. The target of firms is:

$$p_t^* = p_t + \tfrac{\gamma+\psi}{1+\psi\theta}\, x_t,$$

and the price level $p_t$ is the average of the firms' estimates of this target. The remaining equilibrium conditions are:

$$\begin{aligned}
\pi_t &= p_t - p_{t-1},\\
x_t &= E_t x_{t+1} - \gamma^{-1}\left(i_t - E_t\pi_{t+1}\right) + E_t y^n_{t+1} - y^n_t,\\
y^n_t &= \tfrac{1+\psi}{\gamma+\psi}\, a_t,\\
y_t &= x_t + y^n_t,\\
i_t &= \rho_r\, i_{t-1} + (1-\rho_r)\left(\phi_\pi\pi_t + \phi_x x_t + \phi_{\Delta y}\Delta y_t\right) + u_t,\\
a_t &= \rho_a a_{t-1} + \sigma_a\varepsilon^a_t,\\
u_t &= \sigma_u\varepsilon^u_t,
\end{aligned}$$

with $x_t$ the output gap and $y^n_t$ natural output. Convergence is measured on $p_t$ and $x_t$.

[18] The toolbox fits a VARMA($p$, $p-1$) by least squares on the impulse responses, choosing the lowest $p$ between 2 and 10 that fits within a tolerance, and keeping it across iterations while the fit stays within it; unit roots are fitted in first differences. When no $p$ fits every sequence to within a tenth of a percent of its sum of squares, the compact state is not used. In the replications of Section 5, the orders are fixed.

**B.2. Maćkowiak-Wiederholt 2015.** Firms set prices, and households set consumption and their wage, under rational inattention; prices and wages are otherwise flexible. The targets of firms, for prices, and of households, for consumption and the wage, are:

$$\begin{aligned} p_t^* &= p_t + \tfrac{mc_t}{1+\theta(1-\varphi)/\varphi}, \\ c_t^* &= c_{t+1}^* - \gamma^{-1}\left(i_t - \pi_{t+1}\right), \\ w_t^* &= \tfrac{\gamma c_t^* + \psi n_t + \psi\theta_w w_t}{1+\theta_w\psi}, \end{aligned}$$

and the price level $p_t$, consumption $c_t$ and the real wage $w_t$ are the averages of the agents' estimates of these targets. The remaining equilibrium conditions are:

$$\begin{aligned} y_t &= a_t + \varphi n_t, \\ mc_t &= w_t + (1-\varphi)n_t - a_t, \\ \pi_t &= p_t - p_{t-1}, \\ i_t &= \rho_r i_{t-1} + (1-\rho_r)\left(\phi_\pi \pi_t + \phi_x(y_t - y_t^n)\right) + u_t, \\ y_t^n &= \tfrac{1+\psi}{\varphi\gamma+\psi+1-\varphi}\, a_t, \\ y_t &= c_t, \\ a_t &= \rho_a a_{t-1} + \sigma_a \varepsilon_t^a, \\ u_t &= \rho_u u_{t-1} + \sigma_u \varepsilon_t^u, \end{aligned}$$

with $n_t$ hours worked and $mc_t$ the real marginal cost. Convergence is measured on $p_t$, $mc_t$, $i_t$, $\pi_t$, $n_t$ and $w_t$.

**B.3. Maćkowiak-Wiederholt 2025.** Firms choose capital and labor, and households choose consumption and labor supply, under rational inattention; the households' budget constraint ties consumption to labor income and dividends, so their attention problem bears on labor supply alone. The targets of firms, for capital and labor demand, and of households, for labor supply, are:

$$\begin{aligned} k_t^* &= \tfrac{E_t a_{t+1} - \varphi w_{t+1} - \frac{1-\varphi}{1-\beta(1-\delta)}\, r_t}{1-\alpha-\varphi}, \\ l_t^{d*} &= \tfrac{a_t - w_t}{1-\varphi}, \\ l_t^{s*} &= w_t - \gamma\left(\omega_W w_t + \omega_D d_t\right), \end{aligned}$$

and capital $k_t$, the composite $l_t - \frac{\alpha}{1-\varphi}k_{t-1}$ and scaled labor supply $(\gamma\omega_W + \psi)\, l_t$ are the averages of the agents' estimates of these targets. The remaining equilibrium

conditions are:

$$\begin{aligned} y_t &= a_t + \alpha k_{t-1} + \varphi l_t, \\ inv_t &= \tfrac{k_t - (1-\delta)k_{t-1}}{\delta}, \\ d_t &= \tfrac{y_t - \varphi(w_t + l_t) - I_Y\, inv_t}{D_Y}, \\ c_t &= \omega_W (w_t + l_t) + \omega_D d_t, \\ r_t &= \gamma\left(c_{t+1} - c_t\right), \\ a_t &= \rho_a a_{t-1} + \sigma_a \varepsilon^a_{t-h}, \end{aligned}$$

with $inv_t$ investment, $d_t$ dividends, $I_Y = \alpha\beta\delta/(1-\beta(1-\delta))$, $D_Y = 1-\varphi-I_Y$, $\omega_W = \varphi/(1-I_Y)$ and $\omega_D = D_Y/(1-I_Y)$. The technology innovation is announced $h$ periods in advance; $h = 0$ is the standard technology shock and $h > 0$ a news shock. As in Maćkowiak and Wiederholt (2025), the benefit of information is not discounted in the attention problems. Convergence is measured on $w_t$, $r_t$ and $d_t$.

**B.4. Calibratons.**

## C. Contraction Rates of the Outer Iterations

This appendix defines the contraction rate $\rho$ and the feedback $\mu$ reported in Table 2, and explains how we estimate them.

**Damped iteration.** Both procedures are damped fixed-point iterations (Judd, 1998, sec. 3.9). Let $\boldsymbol{z}$ denote what a procedure updates: the expectation matrices $\boldsymbol{\mathcal{E}}$ in ours, the aggregate paths $\boldsymbol{x}$ in the *tâtonnement*. One full iteration maps $\boldsymbol{z}$ into $G(\boldsymbol{z})$, and the solution is the fixed point $\boldsymbol{z}^* = G(\boldsymbol{z}^*)$. With a step $\omega > 0$, each iteration moves only part of the way toward the new value ($\omega < 1$) or beyond it ($\omega > 1$):

$$\boldsymbol{z}^{(n+1)} = (1-\omega)\,\boldsymbol{z}^{(n)} + \omega\, G\big(\boldsymbol{z}^{(n)}\big).$$

**Contraction rate and feedback.** Near the fixed point, $G$ is approximately linear, $G(\boldsymbol{z}) - \boldsymbol{z}^* \approx \boldsymbol{J}(\boldsymbol{z} - \boldsymbol{z}^*)$, where $\boldsymbol{J}$ is the derivative of $G$ at $\boldsymbol{z}^*$, so the error $\boldsymbol{e}^{(n)} = \boldsymbol{z}^{(n)} - \boldsymbol{z}^*$ evolves as $\boldsymbol{e}^{(n+1)} = [(1-\omega)\boldsymbol{I} + \omega\boldsymbol{J}]\,\boldsymbol{e}^{(n)}$. An eigenvalue $\mu_i$ of $\boldsymbol{J}$ thus becomes $1-\omega(1-\mu_i)$, and once the faster directions have died out, the error shrinks each iteration by the factor

$$\rho = \max_i \big|1 - \omega(1-\mu_i)\big|,$$

the *contraction rate*. The iteration converges if $\rho < 1$, and needs about $\log(\eta/d_0)/\log\rho$ iterations to bring a distance $d_0$ below a tolerance $\eta$. We call the eigenvalue $\mu$ that

Table 5: Calibrations

| | | | | MW25 | |
|---|---|---|---|---|---|
| | | AY21 | MW15 | $h = 0$ | $h = 2$ |
| $\beta$ | Discount factor | 0.99 | 0.99 | 0.99 | 0.99 |
| $\gamma$ | Inverse IES | 2.5 | 1 | 1 | 1 |
| $\psi$ | Inverse Frisch elasticity | 2.5 | 0 | $10^{-5}$ | $10^{-5}$ |
| $\theta$ | Elasticity of demand | 10 | 4 | – | – |
| $\theta_w$ | Elasticity of substitution between labor types | – | 4 | – | – |
| $\varphi$ | Labor elasticity | – | 2/3 | 0.65 | 0.65 |
| $\alpha$ | Capital elasticity | – | – | 0.33 | 0.33 |
| $\delta$ | Depreciation rate | – | – | 0.025 | 0.025 |
| $\lambda_f$ | Firms' cost of attention | 1.546 | $2.89 \times 10^{-4}$ | $6 \times 10^{-5}$ | $6.2 \times 10^{-5}$ |
| $\lambda_h$ | Households' cost of attention | – | $2.89 \times 10^{-4}$ | $1 \times 10^{-5}$ | $3 \times 10^{-5}$ |
| $\phi_\pi$ | Taylor rule, inflation | 2.028 | 1.5 | – | – |
| $\phi_x$ | Taylor rule, output gap | 0.168 | 0.125 | – | – |
| $\phi_{\Delta y}$ | Taylor rule, output growth | 3.122 | – | – | – |
| $\rho_r$ | Interest rate smoothing | 0.9457 | 0.85 | – | – |
| $\rho_a$ | Persistence of technology | 0.85 | 0.93 | 0.9 | 0.9 |
| $\sigma_a$ | Std. of technology innovations | 1.56 | 0.64 | 0.008 | 0.008 |
| $\rho_u$ | Persistence of monetary shocks | 0 | 0 | – | – |
| $\sigma_u$ | Std. of monetary innovations | 0.279 | 0.119 | – | – |
| $T$ | Truncation horizon | 200 | 250 | 200 | 200 |

*Notes:* Costs of attention are per nat in AY21 and per bit in MW15 and MW25. In MW25, $h$ is the news horizon. In MW15, both costs are half the published value: at the published value our solution is inconsistent with theirs, likely because the authors solve the attention problem through numerical optimization and truncate its horizon.

attains the maximum at step $\omega$ the *feedback*: the share of an error in that direction that survives one full, undamped iteration. Each eigenvalue $\mu_i$ is independent of the step, but the direction attaining the maximum can change with it, so the feedback we report describes the dominant direction at the step used rather than all directions.

The feedback determines which steps work. With real eigenvalues below one, direction $i$ converges if and only if $\omega < 2/(1-\mu_i)$, and a single step must satisfy this bound for every direction at once. A feedback close to one leaves most of the error after each undamped iteration and calls for a large step, but a step above $2/(1-\mu_i)$ makes any direction with a lower eigenvalue $\mu_i$ diverge. A strongly negative feedback means that a full iteration overshoots the solution and oscillates; only a step below $2/(1-\mu)$ keeps it stable, and with so small a step the directions with eigenvalues

closer to one contract slowly. A feedback above one cannot be fixed by any step.

**Estimation.** We infer $\rho$ and $\mu$ from the distances the iteration produces rather than by computing $\boldsymbol{J}$. At each iteration we record $d_n$, the largest change in the aggregate variables that enter the agents' problems, between the current paths and those implied by the undamped update, the same quantity the stopping rule uses. Near the fixed point, $d_{n+1}/d_n \to \rho$ (Judd, 1998, sec. 2.8). A single ratio is noisy, because the variable attaining the largest change can switch between iterations and the attention problems are solved to a finite tolerance, so we estimate $\rho$ by the geometric mean of the ratios over the last 50 iterations of each stage, $\hat{\rho} = (d_N/d_{N-50})^{1/50}$, and recover $\hat{\mu} = 1-(1-\hat{\rho})/\omega$. This assumes that the slowest direction does not oscillate; if it does, the same $\hat{\rho}$ is also consistent with $\hat{\mu} = 1-(1+\hat{\rho})/\omega$, a large negative number at the small steps of the published procedures. When a stage converges in fewer than 50 iterations, as ours often does, the estimate uses all of them and is less precise. Figure 1 shows the distances $d_n$ themselves.

## D. Likelihood-Based Estimation: Prior and Posterior Distributions

Table 6 reports the priors, and Figures 2 and 3 show the posterior distributions of the two estimations of Section 5.2.

With three parameters, the data are informative about all three: the posteriors are much tighter than the priors, and the 90% intervals contain the true values. With nine parameters, all nine intervals also contain the true values, but the data say little about $\phi_\pi$, $\phi_y$ and $\rho_m$, whose posteriors stay close to their priors, and the interval for the firms' cost of attention is much wider, $[65.8, 222.5]$ against $[92.5, 126.5]$.

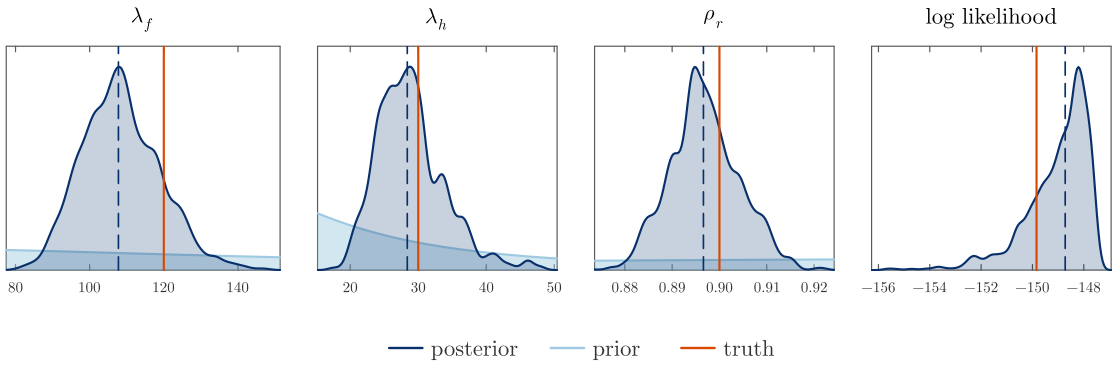
The last panel of each figure shows the distribution of the log likelihood over the posterior draws. Its value at the true parameters, $-149.84$, is unremarkable within this distribution: 21% of the draws have a lower likelihood with three parameters, and 30% with nine. Like a posterior draw, the truth lies below the peak of the likelihood, which is at the parameters that best fit the sample.

Table 6: Priors

| Parameter | Truth | Distribution | Mean | Std. dev. |
|---|---|---|---|---|
| Firms' cost of attention $\lambda_f$ | 120 | Log-normal | 247.3 | 324.2 |
| Households' cost of attention $\lambda_h$ | 30 | Log-normal | 33.0 | 43.2 |
| Interest-rate smoothing $\rho_r$ | 0.9 | Beta | 0.81 | 0.15 |
| Taylor rule, inflation $\phi_\pi$ | 1.5 | Gamma | 1.5 | 0.25 |
| Taylor rule, output $\phi_y$ | 0.125 | Gamma | 0.125 | 0.1 |
| Technology persistence $\rho_a$ | 0.9 | Beta | 0.81 | 0.15 |
| Technology volatility $\sigma_a$ | 0.85 | Inverse gamma | 0.84 | 8.42 |
| Monetary persistence $\rho_m$ | 0.1 | Beta | 0.23 | 0.15 |
| Monetary volatility $\sigma_m$ | 0.2 | Inverse gamma | 0.40 | 3.96 |

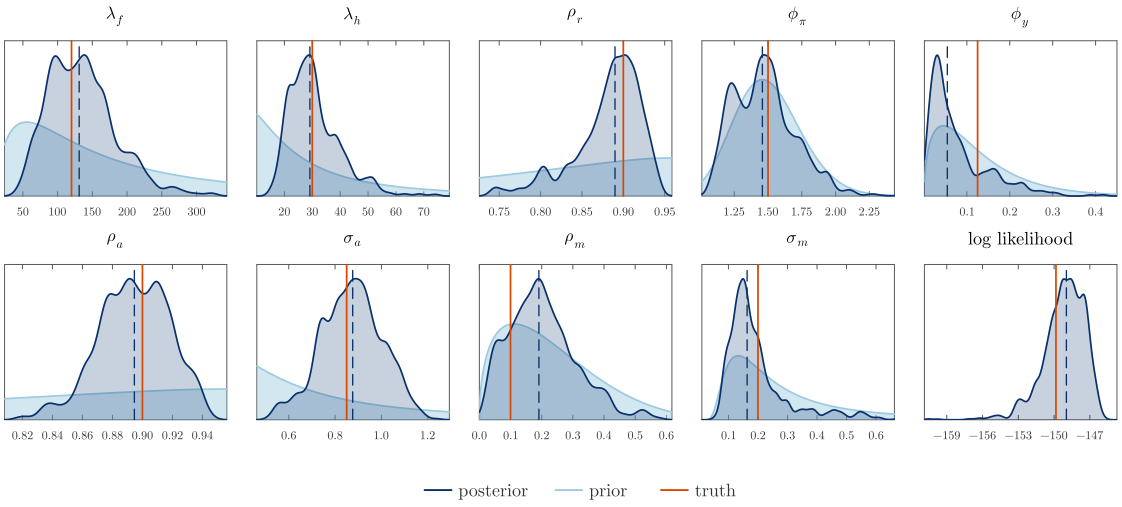
*Notes:* Beta priors on the persistence parameters are truncated at 0.99. Costs of attention are per nat, on responses in percent.

Figure 2: Posterior Distributions, Three Parameters




*Notes:* Kernel densities of the 4,000 draws after burn-in of the random-walk Metropolis–Hastings chain, 200 simulated quarters. The dashed blue lines are the posterior medians. The last panel shows the log likelihood of the draws, and the orange line its value at the true parameters.

Figure 3: Posterior Distributions, Nine Parameters




*Notes:* Kernel densities of the 4,000 draws after burn-in of the random-walk Metropolis–Hastings chain, 200 simulated quarters. The dashed blue lines are the posterior medians. The last panel shows the log likelihood of the draws, and the orange line its value at the true parameters.